

Assignment 4 - Arrays and Methods

Deductions for late submissions is 5%/day, up to a maximum of 3 days including weekends.

FOR ALL PROBLEMS CREATE THE FLOWGORITHM FLOWCHART. Submit 4 .fprg files (called Question1.fprg, Question2.fprg, Question3.fprg, Question4.fprg) zipped onto LEA. As well give your methods meaningful names - make sure they are descriptive.

1. Rainfall Statistics

Design a program that lets the user enter the total rainfall for each month of a year into an array. The program should use methods to calculate and display the total rainfall for the year (do this entirely in the method), another method to compute and return the average monthly rainfall (display the average in your main), and create methods to display the months with the highest and lowest amounts of rain..

2. Charge Account Validation

Design a program that asks the user to enter a charge account number. The program should then call a method to determine whether the number is valid by comparing it to the following list of valid charge account numbers:

5658845 4520125 7895122 8777541 8451277 1302850
8080152 4562555 5552012 5050552 7825877 1250255
1005231 6545231 3852085 7576651 7881200 4581002

The method simply returns a boolean value. These numbers should be stored in an array. Use the sequential search algorithm to locate the number entered by the user. If the number is in the array, the program should display a message indicating the number is valid. If the number is not in the array, the program should display a message indicating the number is invalid.

3. Payroll

Design a program that uses the following parallel arrays:

§ `empId`: An array of seven `Integers` to hold employee identification numbers.

The array should be initialized with the following numbers: 56588 45201 78951
87775 84512 13028 75804

§ `hours`: An array of seven `Integers` to hold the number of hours worked by each employee.

§ `payRate`: An array of seven `doubles` to hold each employee's hourly pay rate.

§ `wages`: An array of seven `doubles` to hold each employee's gross wages.

The program should relate the data in each array through the subscripts. For example, the number in element 0 of the hours array should be the number of hours worked by the employee

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whose identification number is stored in element 0 of the `empld` array. That same employee's pay rate should be stored in element 0 of the `payRate` array.

The program should display each employee number and ask the user to enter that employee's hours and pay rate. It should then calculate the gross wages for that employee (hours times pay rate), which should be stored in the `wages` array. After the data has been entered for all the employees, the program should display each employee's identification number and gross wages. Create another method to display each employee's identification number and gross wages. Question to think about (no you don't have to do anything for the assignment) – Do we really

4. Driver's License Exam

The local Driver's License Office has asked you to write a program that grades the written portion of the driver's license exam. The exam has 10 multiple choice questions. Here are the correct answers:

1. B 2. D 3. A 4. A 5. C 6. A 7. B 8. A 9. C 10. D

You are to use methods for this question. You decide how many methods you would need. Your program should read the correct answers from an array. It should ask the user to enter the student's answers for each of the 10 questions, and the answers should be stored in another array. After the student's answers have been entered, the program should display a message indicating whether the student passed or failed the exam. (A student must correctly answer at least 6 of the 10 questions to pass the exam.)

After all the answers have been inputted. A report should display:

- A list showing the question number with the correct answer, entered answer and Correct/Incorrect result.
- The total number of correctly answered questions
- The total number of incorrectly answered questions Example:

Question	Answer	User input	Result
1.	B	A	Incorrect
2.	D	D	Correct
3.	A	A	Correct
...			
10.	D	A	Incorrect

Total number of correct answers: 6

Total number of incorrect answers: 4 Result: PASS

Input Validation: Only accept the letters A, B, C, or D as answers and an error message if anything else is entered.